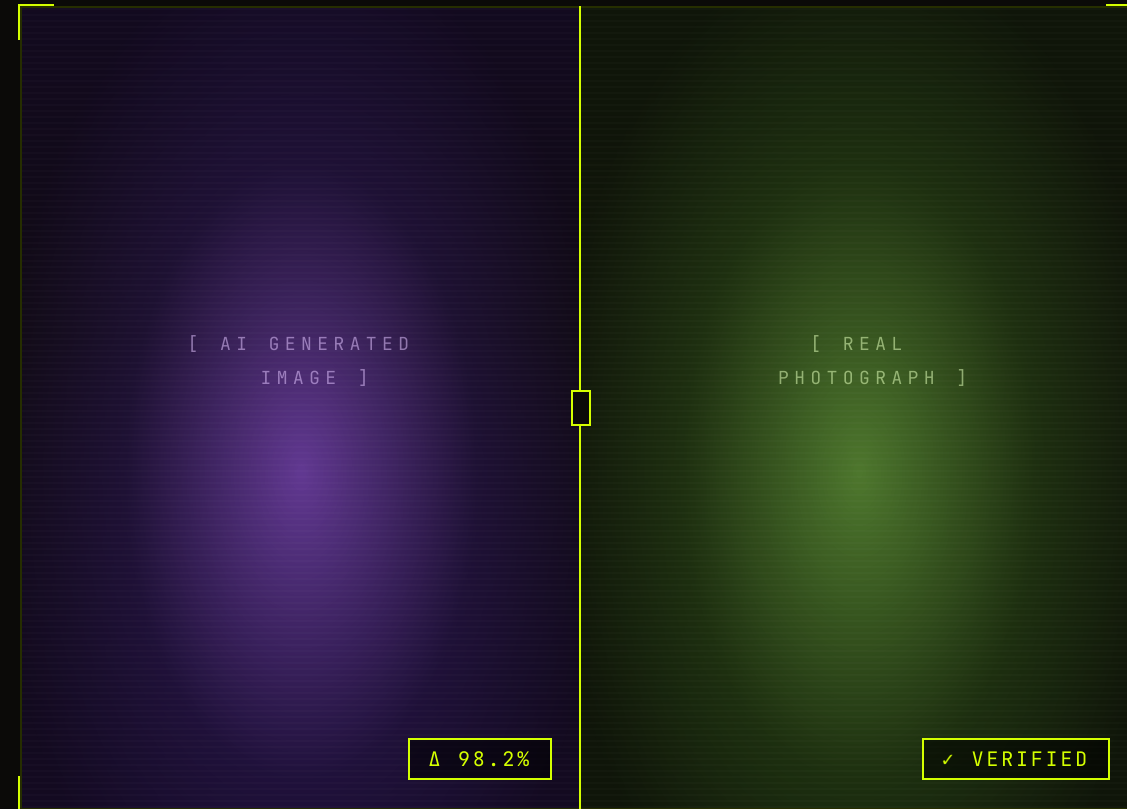


X 0412 · Y 0288 · Δ 98.2%

AUTHENTICITY REPORT

Is this image *real?*



● AI GENERATED

● REAL IMAGE

An MLOps pipeline for AI-generated vs. real image classification.

§ 01 GOAL & MOTIVATION

MLOps pipeline for AI-generated vs. real image classification.

CONTEXT

Generative models produce increasingly photorealistic images. Authenticity detection is a useful and well-scoped task that lets us focus on the **operations** layer of ML.

IN SCOPE

Data versioning · experiment tracking · model registry · CI automation · containerized serving.

GOAL

Build a reproducible MLOps pipeline that classifies an input image as **real** or **ai-generated** with a confidence score.

Every model is pinned to a *dataset version*.

SOURCE

AI Generated Images vs. Real Images

kaggle.com/datasets/tristanzhang32/ai-generated-images-vs-real-images

FAKE	30,000	REAL	30,000
Stable Diffusion	10,000	Pexels / Unsplash	22,500
MidJourney	10,000	WikiArt	7,500
DALL-E	10,000		

labels	real / fake
task	binary classification
splits	train / val / test
versioning	DVC
storage	external remote
license	academic use



§ 03 PIPELINE / FLOW MAP

01  <i>Kaggle</i> ingest · raw/	02  <i>DVC</i> versioned · hashed	03  <i>PyTorch</i> train · EfficientNet-B0	04  <i>MLflow</i> metrics · registry
05  <i>GH Actions</i> CI · on push(main)	06  <i>ONNX</i> export · portable	07  <i>FastAPI + Docker</i> serve · /predict	08  <i>Streamlit</i> demo · UI

§ 04 COMPONENTS / ENTERED INTO EVIDENCE

 <i>DVC</i> DATA VERSIONING Version-controls datasets via hashes and links them to Git commits through metadata pointers.	 <i>Git</i> SOURCE CONTROL Source control for code; commit SHA used by CI/MLflow to trace experiments.	 <i>PyTorch</i> TRAINING EfficientNet-B0 fine-tune.	 <i>MLflow</i> TRACK & REGISTER Experiment tracking and model registry (metrics, parameters, artifacts, model lifecycle stages).
 <i>GH Actions</i> CI AUTOMATION CI pipeline to trigger training, evaluation, and optional model registration on changes.	 <i>ONNX</i> PORTABLE EXPORT Portable model format enabling cross-framework inference via runtimes; decouples training from serving.	 <i>FastAPI</i> REST SERVING Lightweight REST API layer for serving ML models (e.g., ONNX).	 <i>Docker</i> CONTAINERIZATION Containerized runtime ensuring consistent deployment across environments.
 <i>Streamlit</i> DEMO UI Lightweight UI for interactive inference and model demonstration.	 <i>GitHub</i> COLLABORATION Code, PRs, workflows.	 <i>Kaggle</i> DATA SOURCE AI vs real image dataset.	[END OF LIST]

§ 05 WHY THIS IS MLOPS — NOT JUST ML

Any historical run can be *rebuilt*.

- 01 — **Reproducibility** — rebuild any past run from commit + DVC hash.
- 02 — **Traceability** — MLflow links metrics ↔ params ↔ artifacts ↔ data.
- 03 — **Automation** — CI orchestrates training, evaluation, and optional model registration based on triggers.
- 04 — **Deployability** — ONNX + Docker make the model portable.
- 05 — **Iterability** — new data → re-version → retrain → re-register.

EVERY MODEL = (*code version* · *data-version* · *experiment run*) — if any one is missing, the model is not reproducible.